02/08/2026

Code-E

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 AIATS Prime for NEET-2027(XII Passed)_Unit Test-01 **Time : 180 Min.**

PHYSICS

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CHEMISTRY

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BOTANY

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ZOOLOGY

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175. (4)
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177. (2)
178. (3)
179. (1)
180. (3)

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Hints and Solutions

PHYSICS

(1) **Answer :** (1)
Hint:
The difference with average value should be least.
Solution:
Δt_1 = |10.00 – 9.95| = 0.05 s
Δt_2 = |10.00 – 10.10| = 0.10 s

(2) **Answer :** (2)
Solution:
0.03708 → 4 significant figures
902.60 → 5 significant figures
5.00×10^{-2} → 3 significant figures

(3) **Answer :** (2)
Solution:
$\sigma = \frac{F}{\pi r^2}$
Maximum percentuge error in $\sigma(\%\ \sigma) = 1 \times \%F + 2 \times \%r$
$\%F = \frac{0.005}{5} \times 100 = 0.1\%$
$\%r = \frac{0.01}{2} \times 100 = 0\ .5\%$
$\therefore\ \%P = 0.1 + 2(0.5) = 1.1\%$

(4) **Answer :** (4)
Hint:
Dimensional formula of gravitational potential energy is same as that of kinetic energy.
Solution:
$[K] = \left[\frac{1}{2}MV^2\right] = [U]$
$[K] = [MV^2T^0]$

(5) **Answer :** (2)
Hint:
Use rules of significant figures
Solution:
The number 4.700 has four significant figures.
Changing unit does not change the number of significant figures.

(6) **Answer :** (3)
Hint:
Use equation $\frac{dQ}{dt} = eA\sigma T^4$ to find the unit of Stefan's constant (σ)
Solution:
$\frac{dQ}{dt} = eA\sigma T^4$
$\sigma = \left(\frac{dQ}{dt}\right)\left(\frac{1}{eAT^4}\right)$
Unit of σ is $\frac{W}{m^2 K^4}$

(7) **Answer :** (3)
Solution:
Measured value 1.56 is more precise than 1.5
More accurate measurement is not always more precise.

(8) **Answer :** (3)
Solution:
$\omega = 2\pi f$

(9) **Answer :** (1)
Solution:
Angle have unit but no dimensions.

(10) **Answer :** (2)
Solution:

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$[v] = [a]\,[t] \Rightarrow [a] = [LT^{-2}]$ and $[c] = [T]$

$\Rightarrow [a/c] = [LT^{-3}]$

(11) **Answer :** (2)
Solution:
Volume $V = l^3 = 2.5 \times 2.5 \times 2.5$
$V = 15.625\ m^3$
According to rule of significant figure = $16\ m^3$

(12) **Answer :** (2)
Solution:
If 86.45 is rounded off upto three significant figure it becomes 86.4

(13) **Answer :** (2)
Solution:
Percentage error in measurement
$= \frac{\Delta t}{t} \times 100 = \frac{0.2}{200} \times 100 = 0.1\%$

(14) **Answer :** (2)
Hint:
Reading = MSR + VSR
Solution:
Main scale reading, MSR = 6.8 cm
Vernier scale reading, VSR = 4 × LC = 0.04 cm
∴ Reading = 6.8 + 0.04 = 6.84 cm

(15) **Answer :** (2)
Solution:
0.0007 → 1
2.64×10^{24} → 3
6.320 → 4
0.63200 → 5

(16) **Answer :** (4)
Solution:
$\mathrm{L \cdot C} = \frac{1}{10} = 0.1\ \mathrm{mm}$
Zero error = +7 × 0.1 = 0.7 mm
Zero correction = – 0.7 mm

(17) **Answer :** (2)
Solution:
$n_1u_1 = n_1u_2$
$n_1\left[MLT^{-2}\right] = n_2\left[\frac{3M3L}{3^2T^2}\right]$
$n_1 = n_2$

(18) **Answer :** (2)
Solution:
n MSD = $(n + 1)$ VSD
$1\ VSD = \frac{n}{n+1}\ MSD$
$L \cdot C = \left[1 - \frac{n}{n+1}\right](x)$
$L \cdot C = \frac{x}{n+1}$

(19) **Answer :** (2)
Solution:
LC = 1 msd/number of circular scale divisions
= 1 mm/100 = 0.01 mm

(20) **Answer :** (1)
Solution:
Main scale reading = 6 mm
Circular scale reading = 41 × 0.01 = 0.41 mm
Final reading = 6.41 mm

(21) **Answer :** (2)
Solution:

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$L.C. = \frac{\text{Pitch}}{\text{C.S.D}} = \frac{1}{100}\text{ mm} = 0.01\text{ mm}$

Observed diameter = 1 mm + 39 × 0.01 mm = 1.39 mm
Corrected diameter of wire = 1.39 mm + 0.05 mm = 1.44 mm

(22) Answer : (1)
Solution:

$\vec{v}_{m/t} = \vec{v}_{m/g} - \vec{v}_{t/g}$

$\vec{v}_{m/g} = \vec{v}_{m/t} + \vec{v}_{t/g} = 2\hat{i} + 10\hat{i}$

$\vec{v}_{m/g} = (12\hat{i})\text{m/s}$

(23) Answer : (2)
Solution:

$a = 6t + 5$

$\frac{d^2S}{dt^2} = 6t + 5$

$\frac{dS}{dt} = 3t^2 + 5t$

$S = \int_0^4 (3t^2 + 5t)\,dt = 104\text{ m}$

(24) Answer : (2)
Hint:
Distance = $\int \text{speed}\ \ \text{dt}$
Solution:
Distance = $\int_0^6 \text{speed}\ \times\ dt$
Particle direction changes at t = 4 sec

$\text{distance} = \int_0^4 (4t - t^2)\ dt + \int_4^6 (t^2 - 4t)\ dt$

$= \int_0^4 (4t - t^2)\,dt + \int_4^6 (t^2 - 4t)\ dt$

$= \left(2t^2 - \frac{t^3}{3}\right)\Big|_0^4 + \left(\frac{t^3}{3} - 2t^2\right)\Big|_4^6$

$= \frac{32}{3} + \frac{32}{3} = 21.33\text{ (m)}$

(25) Answer : (4)
Hint:
Slope of position – time graph is velocity
Solution:
Velocity is negative and constant till 0 to mid of t_1 and t_2. Velocity is positive from mid of t_1 and t_2 to t_3.

(26) Answer : (1)
Solution:

$\text{a} = \frac{5-10}{5} = -1\text{ m/s}^2$

Retardation = 1 m/s^2

(27) Answer : (4)
Solution:
∵ | Displacement | ≤ Distance
∴ | Average velocity | ≤ Average speed
If displacement is zero, distance may or may not be zero.

(28) Answer : (1)
Solution:

Distance = Area under curve = $\frac{1}{2} \times 5 \times 6 = 15\text{ m}$

(29) Answer : (1)
Solution:

$t_1 = \sqrt{\frac{2h}{g}}$

$t_2 = \sqrt{\frac{2(3h)}{g}} - \sqrt{\frac{2h}{g}}$

$= \left(\sqrt{3} - 1\right) t_1$

$= \left(\sqrt{3} - 1\right) \times 4 \approx 3\ s$

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(30) Answer : (4)
Hint:
Use $v^2 - u^2 = 2as$
Solution:
$0 - 10^2 = 2(-2) \times 5\ s$
$s_1 = \frac{100}{4} = 25$ m
Total distance $= 2s_1 = 25 \times 2$
$= 50$ m

(31) Answer : (3)
Hint:
Change in velocity = Area under *a-t* graph
Solution:
$\Delta v = \int a dt$ = Area under *a-t* graph
For $v_f = v_i \Rightarrow \Delta v = 0$
$\therefore$ Area under *a-t* graph should be zero.
From graph, at $t = 12$ s upper and lower area would become equal in magnitude.

(32) Answer : (2)
Hint:
Average speed = $\frac{\text{Total distance}}{\text{Total time}}$
Solution:
$< v >= \frac{s}{t_1+t_2} = \frac{\frac{s}{2}+\frac{s}{2}}{\frac{s}{2u}+\frac{s}{2v}} = \frac{2uv}{u+v}$

(33) Answer : (3)
Solution:
$v^2 = u^2 + 2gh$
$v^2 = 0 + 2 \times 9.8 \times 4.9$
$\boxed{v = 9.8\ m/s}$

(34) Answer : (3)
Solution:
Area under *v-t* curve = $\int v dt = \Delta x$
Hence area under *v-t* curve gives displacement of the particle.

(35) Answer : (2)
Solution:
For uniformly accelerated motion in straight line
$v_{avg} = \frac{u+v}{2}$
$v_{avg} = \frac{0+(0+at)}{2} = \frac{at}{2}$

(36) Answer : (1)
Solution:
$\text{Average speed, } v_{av} = \frac{\text{Total distance}}{\text{Total time}}$
$v_{av} = \frac{\text{Area under } (v-t) \text{ graph}}{\text{Total time}} = \frac{10+10+10+20}{8}$
$v_{av} = \frac{50}{8} = \frac{25}{4}\ \text{m/s}$
$\text{Average velocity, } \vec{v}_{av} = \frac{\text{Area with sign}}{\text{Total time}} = \frac{10+10-10-20}{8}$
$= \frac{-10}{8} = \frac{-5}{4}\ \text{m}/s$
Hence $\boxed{\begin{array}{l}\text{average speed} = \frac{25}{4}\ \text{m/s} \\ \text{averge velocity} = \frac{-5}{4}\ \text{m/s}\end{array}}$

(37) Answer : (1)
Solution:
$v_{avg} = \frac{x+x}{\frac{x}{v_1}+\frac{x}{v_2}} = \frac{2v_1v_2}{v_1+v_2}$
$= \frac{2v \times 3v}{4v} = \frac{3}{2}v$

(38) Answer : (1)
Hint:

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Use $v = u + gt$ and $s = ut + \frac{1}{2}gt^2$

Solution:

Distance travelled in 2 s while falling freely.

$s = ut + \frac{1}{2}gt^2$

$s_1 = \frac{1}{2} \times 10 \times (2)^2 = 20 \text{ m}$

Velocity at the end of 2 s is

$v = u + gt$

10(2) = 20 m/s

For remaining 60 m, time taken

$t = \frac{60}{20} = 3\text{s}$

Hence total time taken to reach ground = 2 s + 3 s = 5 s.

(39) **Answer :** (1)

Solution:

$s = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 1^2$

= 5 m

(40) **Answer :** (3)

Solution:

Average acceleration

$a_{avg} = \frac{a_1t_1 + a_2t_2}{t_1 + t_2} = \frac{3 \times 2 + 4 \times 3}{2 + 3}$

= 3.6 m s^{-2}

(41) **Answer :** (1)

Solution:

$v = \frac{dx}{dt} = 12t^2 - 3$

$\Rightarrow v = 0$ at $t = 0.5$ s

$\Rightarrow \quad a = \frac{dv}{dt} = 24\ t$

$\Rightarrow$ at $t = 0.5$ s, $a = 12$ m/s^2

(42) **Answer :** (4)

Solution:

Effective acceleration = $g - a = 5$ m/s^2

$u = 0 \ \Rightarrow \ 2 = \frac{1}{2} \times 5 \times t^2$

$\Rightarrow \ t = \frac{2}{\sqrt{5}}$ s

(43) **Answer :** (4)

Solution:

Separation = Relative velocity × time

60 = (16 – 10) × t $\quad \therefore t = 10$ seconds

(44) **Answer :** (3)

Solution:

$\text{t} = \frac{\text{S}_{\text{relative}}}{\text{V}_{\text{relative}}} = \frac{360}{(50+10)} = 6 \text{ s}$

(45) **Answer :** (4)

Solution:

$t = \frac{t_1 t_2}{t_1 + t_2}$

$= \frac{60 \times 30}{60 + 30} = \frac{60 \times 30}{90} = 20\ s$

CHEMISTRY

(46) **Answer :** (2)

Solution:

The positive charge and most of the mass of the atom is concentrated in a small space called nucleus.

(47) **Answer :** (3)

Solution:

Cathode rays originate from cathode.

The charge to mass ratio of cathode rays is independent of gas present in cathode ray tube.

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(48) **Answer :** (2)
Solution:
For the longest wavelength in Lyman series
⇒Transition of electron occurs from $n = 2$ to $n = 1$
$\frac{1}{\lambda} = R_H \times Z^2 \left(\frac{1}{1^2} - \frac{1}{2^2}\right)$
$= R_H \times 1^2 \times \frac{3}{4} = \frac{3R_H}{4}$

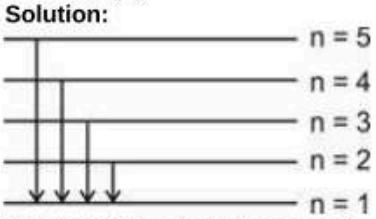
(49) **Answer :** (4)
Solution:
Atoms with same mass number but different atomic number are isobar.

(50) **Answer :** (3)
Hint:
Electromagnetic radiations posses both wave like and particle like properties.
Solution:
• Electromagnetic waves do not require medium and can move in vacuum.
• Radio waves have longer wavelength than that of radio waves.

(51) **Answer :** (3)
Solution:
On bombarding α-particle on thin sheet of beryllium neutron are obtained.

(52) **Answer :** (4)
Solution:
For Li^{2+}
$r_2 = a\frac{n^2}{Z}$
$x = a \times \frac{(2)^2}{3}$
$\frac{3x}{4} = a$
For He^+
$r_1 = \frac{3}{4}x \times \frac{(1)^2}{2} = \frac{3}{8}x$ Å

(53) **Answer :** (3)
Hint:
Number of protons = Atomic number
Number of neutrons = A – Z
Solution:
Number of protons = 30
Number of neutrons = 65 – 30 = 35
Number of electrons = 30 – 2 = 28

(54) **Answer :** (3)
Solution:


Number of lines in UV region = 4.

(55) **Answer :** (4)
Hint:
Power = rate of consumption of energy
Solution:
10 W = 10 Js^{-1}
Energy of photons = $\frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{400 \times 10^{-9}}$
$= 4.9 \times 10^{-19}$
Rate of emission of photon
$= \frac{10}{4.9 \times 10^{-19}} = 2.04 \times 10^{19}$

(56) **Answer :** (2)
Solution:
$E = N_A \frac{hc}{\lambda}$

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$E = \frac{6.02\times10^{23}\times6.626\times10^{-34}\times3\times10^{8}}{6.02\times10^{-7}}$

$E = 19.878 \times 10^4$ J

$E = 198.78 \times 10^3$ J = 198.78 kJ

(57) **Answer :** (1)

Solution:

$h\upsilon = w_0 + KE$

$\frac{hc}{\lambda} = w_0 + 2$ ...(1)

$\frac{4hc}{\lambda} = w_0 + 10$ ...(2)

$4(w_0 + 2) = w_0 + 10$

$3w_0 = 2$

$w_0 = \frac{2}{3}$ eV.

(58) **Answer :** (4)

Hint:

$E_n = \frac{E_1(H)}{n^2} \times Z^2$

Solution:

$E_4(H) = \frac{E_1(H)}{16} = -82 \Rightarrow E_1(H) = -1312 \text{ kJ mol}^{-1}$

$E_2(Li^{2+}) = -\frac{1312}{4} \times 9 = -2952 \text{ kJ mol}^{-1}$

(59) **Answer :** (3)

Solution:

$2\pi r = n\lambda$

$r = \frac{a_o n^2}{Z}$

$\frac{2\pi a_o n^2}{nxZ} = \lambda$

$\frac{2\pi a_o n}{Z} = 3\pi a_o$

$\frac{n}{Z} = 1.5$

(60) **Answer :** (2)

Solution:

Orbital	*n*	*l*
3*p*	3	1
3*d*	3	2
2*p*	2	1
1*s*	1	0

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(61) **Answer :** (3)

Solution:

For H-like species, energy of orbitals increases as value of n increases

(62) **Answer :** (1)

Solution:

For Balmer series, $n_1 = 2,\ n_2 = 3,4,5,\ldots$

for shortest wavelength (λ_1), $n_1 = 2,\ n_2 = \infty$

$\therefore \frac{1}{\lambda_1} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\frac{1}{\lambda_1} = R\times(1)^2\left[\frac{1}{2^2} - \frac{1}{\infty^2}\right] = R\left[\frac{1}{4} - 0\right]$

$\Rightarrow \lambda_1 = \frac{4}{R} \Rightarrow R = \frac{4}{\lambda_1}$

for Li^{2+} ion ($Z = 3$), longest wavelength in Paschen series, $n_1 = 3,\ n_2 = 4$

$\Rightarrow \frac{1}{\lambda_{Li^{2+}}} = R\times(Z)^2\left[\frac{1}{3^2} - \frac{1}{4^2}\right]$

$\Rightarrow \frac{1}{\lambda_{Li^{2+}}} = R\times(Z)^2\left[\frac{1}{9} - \frac{1}{16}\right]$

$\Rightarrow \frac{1}{\lambda_{Li^{2+}}} = R\times\left[1 - \frac{9}{16}\right] = \frac{7R}{16}$

$\Rightarrow \lambda_{Li^{2+}} = \frac{4\lambda_1}{7}$

(63) **Answer :** (3)

Solution:

$_{29}Cu = [Ar]3d^{10}4s^1$

$\therefore$ Orbital angular momentum $= \sqrt{l(l+1)}\ \hbar = 0$

(64) **Answer :** (2)

Hint:

$\lambda = \frac{h}{mv}$

Solution:

$\lambda = \frac{h}{mv} = \frac{6\times10^{-34}}{9\times10^{-31}\times10^{5}}$

$= 6.66 \times 10^{-9}$ m = 6.66 nm

(65) **Answer :** (4)

Solution:

For the 2 electrons in same orbital the value of n, l and m_ℓ are same so these can be distinguished by only spin quantum number.

(66) **Answer :** (4)

Solution:

Law of multiple proportions is valid for those compounds that have same constituent elements.

(67) **Answer :** (1)

Solution:

- $(2l+1)$ represents the number of orbitals in each subshell.
- m_l designates the orientation of the orbital

(68) **Answer :** (2)

Hint:

1 P^{3-} ion has 18 electrons.

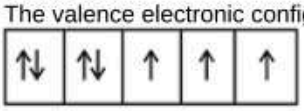
Solution:

1 mol P^{3-} ions have 18 N_A electrons

(69) **Answer :** (4)

Solution:

The valence electronic configuration of Co^{2+} ion is $3d^7$.



So, the number of unpaired electrons is 3.

(70) **Answer :** (2)

Solution:

$2H_2(g) + O_2(g) \rightarrow 2H_2O$

2 Volume 1 Volume

(71) **Answer :** (1)

Solution:

Multiple Prefix

10^{15} peta

10^{12} tera

10^{9} giga

10^{6} mega

(72) **Answer :** (4)

Solution:

Zeros preceding to first non-zero digit are not significant figure will have 2 significant.

(73) **Answer :** (4)

Hint:

Photoelectric effect is the instantaneous emission of electrons from the metal surface when a photon of suitable frequency is incident on it.

(74) **Answer :** (4)

Hint:

Mole of molecule/atom = $\frac{\text{Mass of molecule / atom}}{\text{Molar mass of molecule / atom}}$

Solution:

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Mole of He = $\frac{2x}{4}$

Mole of N_2 = $\frac{x}{28}$

$\frac{\text{Mole of He}}{\text{Mole of } N_2} = \frac{\frac{2x}{4}}{\frac{x}{28}} = \frac{2x}{4} \times \frac{28}{x} = 14:1$

(75) Answer : (3)

Solution:

$X\,(\text{av}) = \frac{54\times 70+53\times 20+55\times 10}{100}$

$= \frac{3780+1060+550}{100} = 53.9\text{u}$

(76) Answer : (3)

Solution:

$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$

0.3 mole of $BaCl_2$ combines with 0.2 mole of Na_3PO_4. Therefore, $BaCl_2$ is limiting reactant.

0.3 mole of $BaCl_2$ will produce 0.1 mole of $Ba_3(PO_4)_2$ and 0.6 mole of NaCl.

(77) Answer : (2)

Hint:

Number of moles = $\frac{\text{Given mass (in g)}}{\text{Molar mass}} = \frac{N}{N_A}$

Solution:

Number of molecules left = $\left(\frac{220\times 10^{-3}}{44} \times 6.022 \times 10^{23}\right) - 10^{21}$

$= 3.011 \times 10^{21} - 10^{21}$

$= 2.011 \times 10^{21}$

(78) Answer : (4)

Hint:

Number of atoms = Moles × N_A × Atomicity

Solution:

Let mass of each of S_2 and S_8 taken be x g

S_2	S_8
Moles = $\frac{x}{32\times 2}$	$\frac{x}{32\times 8}$
Molecules = $\frac{x}{32\times 2}\times N_A$	$\frac{x}{32\times 8}\times N_A$

Let mass of each O_2 and O_3 taken be x g

Number of atoms in O_2 molecule $= \frac{x}{16\times 2}N_A \times 2 = \frac{x}{16}N_A$

Number of atoms in O_3 molecule $= \frac{x}{16\times 3}N_A \times 3 = \frac{x}{16}N_A$

(79) Answer : (3)

Solution:

$\text{Density} = \frac{\text{Molar mass}}{\text{Molar volume}} = \frac{46}{22.4} = 2.05\ \text{g/L}$

(80) Answer : (2)

Solution:

Let the atomic mass of B = y u ; A = x u

In B_2A_3,

2y + 3x = mol. mass of B_2A_3 = $\frac{\text{given weight}}{\text{mole}}$

2y + 3x = $\frac{9}{0.05}$ g

In B_2A,

$\therefore$ 2y + x = $\frac{10}{0.1}$ g

Solving x and y

$\left\{\begin{matrix} x = 40 \\ y = 30 \end{matrix}\right\}$

(81) Answer : (1)

Solution:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6\,CO_2 + 6H_2O$

Mole of CO_2 = $\frac{44}{44} = 1$

Mole of glucose required = $\frac{1}{6}$

Mass of glucose required = $\frac{1}{6} \times 180 = 30\ \text{g}$

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(82) **Answer :** (2)
Solution:
$C_2H_6 + \frac{7}{2}O_2 \longrightarrow 2CO_2 + 3H_2O$
Mole of O_2 required $= \frac{7}{2} \times \frac{3}{30} = \frac{7}{20}$
Volume of O_2 required at STP
$= \frac{7}{20} \times 22.4\ \mathrm{L} = 7.84\ \mathrm{L}$

(83) **Answer :** (2)
Solution:
$CaCO_3(s) + 2\,HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + 2H_2O(l)$
2 moles of HCl react with = 100 g of $CaCO_3$
0.025 mole of HCl will react = $\frac{100}{2} \times 0.025$ g $CaCO_3$
= 1.25 g of $CaCO_3$
$x \times \frac{90}{100} = 1.25$
$\frac{12.5}{9} g = 1.39\ g$
Mass of 90% pure $CaCO_3$ will be = 1.39 g

(84) **Answer :** (3)
Hint:
$\mathrm{Molality} = \frac{\mathrm{Number\ of\ moles\ of\ solute}}{\mathrm{Mass\ of\ solvent\ in\ kg}}$
Solution:
• 0.25 molal aqueous solution of urea means that 0.25 moles of urea is dissolved in 1000 g of H_2O
• So mass of urea = 0.25 × 60 = 15 g
• So mass of solution = 1000 +15 = 1015 g = 1.015 kg
∵ 1.015 kg of urea solution contains 15 g of urea
∴ 2.5 kg of urea solution contains $\frac{15}{1.015} \times 2.5$ = 37 g

(85) **Answer :** (4)
Hint:
$M_1V_1 + M_2V_2 = M_3V_3$
Solution:
20 × 1 + 200 × 0.25 = 220 × M
20 + 50 = 220 M
M = $\frac{70}{220}$ = 0.32 M

(86) **Answer :** (3)
Solution:

COMPOUND	MOLECULAR FORMULA	EMPIRICAL FORMULA	MOLECULAR MASS
Acetylene	C_2H_2	CH	26 u
Benzene	C_6H_6	CH	78 u

As , $26 \times 3 = 78$
∴ Molecular mass of $C_2H_2 \times 3$ = Molecular mass of C_6H_6

(87) **Answer :** (3)
Solution:

$$\underset{1\ \mathrm{mol}}{Zn} + 2\,HCl \rightarrow \underset{22.4\ \mathrm{L} = 22400\ \mathrm{ml}}{ZnCl_2 + H_2(g)}$$

22400 ml of H_2 gas is produced from Zn = 65 g
1.12 ml of H_2 gas is produced from Zn = $\frac{65}{22400} \times 1.12$ g = 3.25×10^{-3} g
i.e., 32.5×10^{-4} g

(88) **Answer :** (1)
Solution:

Element	%	Mole	
C	40	$\frac{40}{12} = 3.33$	$\frac{3.33}{3.33} = 1$
H	13.33	$\frac{13.33}{1} = 13.33$	$\frac{13.33}{3.33} = 4$
N	46.67	$\frac{46.67}{14} = 3.33$	$\frac{3.33}{3.33} = 1$

$CH_4N \rightarrow$ Empirical formula

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(89) **Answer :** (2)
Solution:
Cr (Z = 24) = $1s^2 2s^2 2p^6\ 3s^2 3p^6 3d^5 4s^1$
n + l = 4 is for orbitals 4s and 3p
Hence the number of electrons in orbitals having
n + l = 4 is equal to 7.

(90) **Answer :** (4)
Solution:
(a) 4.8 g of O_3: $\frac{4.8}{48} \times N_A \times 3 = 0.3\,N_A$
(b) 3.2 g of SO_2: $\frac{3.2}{64} \times N_A \times 2 = 0.1\,N_A$
(c) 3.3 g of CO_2: $\frac{3.3}{44} \times N_A \times 2 = 0.15\,N_A$
(d) 5.6 g of CO: $\frac{5.6}{28} \times N_A \times 1 = 0.2\,N_A$

BOTANY

(91) **Answer :** (3)
Solution:
The correct sequence of events during prophase I of meiosis is b, d, c, a. i.e., zygotene, pachytene, diplotene, and diakinesis, respectively

(92) **Answer :** (2)
Solution:
Meiosis II takes place in haploid cells, which are produced after meiosis I.
Meiosis II is equational division as the number of chromosomes in the daughter cells remains the same as in the cells produced at the end of meiosis I.

(93) **Answer :** (3)
Solution:
In Quiescent or G_0 stage, cells do not proliferate unless they are called on to do so and this stage is metablically active.

(94) **Answer :** (3)
Hint:
Cell division will occur for eight times.
Solution:
Number of generations $= \frac{160}{20} = 8$
Number of cells = $2^8 \times 10^3$ per ml.
= 256×10^3 cells/ml

(95) **Answer :** (4)
Hint:
Metaphasic plate is formed on the equatorial plane.
Solution:
The metaphase plate is an imaginary plane where centromere of chromosomes align themselves.

(96) **Answer :** (1)
Solution:
Figure A and B represent prophase-II and anaphase-I, respectively. Anaphase promoting complex regulates the cell cycle by facilitating the transition from metaphase to anaphase. During interkinesis, chromosome elongate but do not form chromatin fibres.

(97) **Answer :** (4)
Solution:
Nerve cells in adult animal cell do not divide.

(98) **Answer :** (4)
Solution:
Division of the cytoplasm occurs by formation of furrow in animal cells.

(99) **Answer :** (4)
Hint:
During zygotene, the homologous chromosomes come to lie side by side in pairs and this pairing is called synapsis.
Solution:
In diplotene stage, the synaptonemal complex formed during zygotene dissolves.
The recombination of genetic material that occurs during pachytene is mediated by the enzyme recombinase.

(100) **Answer :** (2)

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Solution:
Its parental cell had sixteen chromosomes which form eight bivalent or tetrad during pachytene of prophase-I during meiosis-I.

(101) Answer : (1)
Solution:
Anaphase II occurs in haploid cells and involves centromere division.

(102) Answer : (3)
Solution:
DNA replication occurs in S phase of cell cycle due to which amount of DNA becomes double but number of chromosomes remains the same..

(103) Answer : (1)
Solution:
Meristem divides throughout the life.

(104) Answer : (1)
Hint:
Interzonal fibres are seen between two separating chromatids.
Solution:
During anaphase of mitosis, the formation of interzonal fibres occur.

(105) Answer : (3)
Solution:
DNA replication occur once in S-phase only.

(106) Answer : (1)
Hint:
This is the phase of actual cell division.
Solution:
M phase is the most dramatic period of the cell cycle, involving a major reorganisation of virtually all components of the cell.

(107) Answer : (2)
Hint:
Meiosis is the reduction division.
Solution:
Fertilisation restores the diploid phase in life cycle of sexually reproducing organisms.

(108) Answer : (2)
Hint:
It is fourth stage of prophase I.
Solution:
In oocytes of some vertebrates, diplotene lasts for month or years. It is called dictyotene stage.

(109) Answer : (2)
Solution:
Centrosomes radiate some microtubules called asters.

(110) Answer : (1)
Solution:
Correct order for cell cycle is
$G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
Hence, G_2 occurs between S and M phase.

(111) Answer : (3)
Solution:
Cell divides itself into two daughter cells by the separation of cytoplasm, at the end of which cell division gets completed.

(112) Answer : (2)
Solution:
Yeast can progress through the cell cycle in only about 90 minutes

(113) Answer : (4)
Solution:
Anything less than a complete structure of a cell does not ensure independent living. Hence, cell is the fundamental structural and functional unit of all living organism.

(114) Answer : (2)
Solution:
Rudolf Virchow modified the hypothesis of Schleiden and Schwann to give cell theory, its final shape.
Cell theory as understood today is :
(i) All living organisms are composed of cell and product of cells.
(ii) All cells arise from pre-existing cells. (Given by Rudolf Virchow)

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(115) Answer : (3)
Hint:
RBCs are round and biconcave.
Solution:
White blood cells – amoeboid.
Mesophyll cells – round and oval.
Nerve cells – branched and long.

(116) Answer : (4)
Solution:
Prokaryotes lack membrane bound organelles.

(117) Answer : (1)
Hint:
Plasmid is a small circular DNA and extranuclear genetic material found in many bacteria.
Solution:
• Plasmid is absent in eukaryotes
• Many ribosomes attach to a single m-RNA to form polysomes.

(118) Answer : (3)
Hint:
Prokaryotic cells show the presence of fimbriae.
Solution:
The middle layer of the cell envelope (cell wall) in the prokaryotic cell prevents these cells from bursting or collapsing.
Glycocalyx differs in composition and thickness in different forms.

(119) Answer : (2)
Solution:
Cell-membranes possess carbohydrates.
Fluid-mosaic model of cell membrane proposed by Singer and Nicolson is widely accepted.

(120) Answer : (2)
Hint:
Almost all plant cells lack centriole.
Solution:
Plant cells and animal cells are different as the former possess cell wall, plastids and a large central vacuole which are absent in the animal cells.

(121) Answer : (3)
Solution:
Secondary cell wall is formed on the inner side (towards the cell membrane) of the primary cell wall.

(122) Answer : (1)
Solution:
Golgi apparatus was first observed and described by Camillo Golgi. Camillo Golgi described Golgi apparatus as a densely stained reticular structures present near the nucleus of the cell. Therefore, these were given the name Golgi body, after his name.

(123) Answer : (3)
Solution:
Telomeres are the ends of chromosomes, which prevent their shortening.

(124) Answer : (4)
Solution:
In animal cells, detoxification of drugs and steroid synthesis are associated to smooth endoplasmic reticulum.

(125) Answer : (2)
Solution:
Golgi complex is involved in glycosylation of proteins.
Contractile vacuole – Excretion
RER – Protein synthesis
Amyloplast – Stores starch

(126) Answer : (2)
Hint:
Hub in centriole is proteinaceous.
Solution:
The central part of the proximal region of the centriole is proteinaceous and called the hub.

(127) Answer : (2)
Solution:
In plants, the tonoplast facilitates the transport of a numbers of ions and other materials against concentration gradients into the vacuole. Hence, their concentration is significantly higher in the vacuole than in the cytoplasm.

(128) Answer : (4)

Hint:
The inner compartment of mitochondria is filled with a dense homogeneous substance called the matrix.
Solution:
The matrix of mitochondria possesses single circular DNA molecule, a few RNA molecules, ribosomes (70S) and the components required for the synthesis of proteins.

(129) Answer : (3)
Solution:
Outer membrane of chloroplast is relatively more permeable and has porins.
Grana is the active site for light reaction.
Stroma is the site for dark reaction or fixation of CO_2 into sugar.

(130) Answer : (2)
Hint:
Secondary constriction is a non-staining region of SAT chromosome.
Solution:
Satellite is a region beyond secondary constriction in SAT chromosome.

(131) Answer : (3)
Hint:
Cell wall prevents the cell from bursting or collapsing.
Solution:
Slime layer protects the cell from loss of water and nutrients. Capsule allows bacterium to hide from host's immune system. Cell membrane in prokaryotes is similar structurally to that of eukaryotes.

(132) Answer : (3)
Solution:
Gas vacuoles are found in blue-green and purple and green photosynthetic bacteria.

(133) Answer : (3)
Solution:
During anaphase, submetacentric chromosomes appear L shaped as centromere is present just away from the middle.

(134) Answer : (4)
Solution:
Packed DNA has RNA, histones and some non-histone proteins.

(135) Answer : (2)
Solution:
Plasmodesmata connects the cytoplasm of two plant cells.

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ZOOLOGY

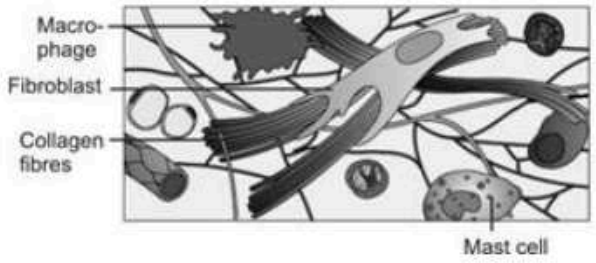
(136) Answer : (1)
Solution:
Lacunae and lamella are present in bone.

(137) Answer : (2)
Solution:

PCT of nephron	–	Brush bordered cuboidal epithelium
Bronchioles	–	Ciliated columnar epithelium
Stomach and intestine	–	Columnar epithelium
Wall of blood vessels	–	Simple squamous epithelium

(138) Answer : (1)
Solution:
Tendons and ligaments are dense regular white fibrous connective tissue in the form of cord.

(139) Answer : (1)
Hint:
Areolar tissue
Solution:



(140) Answer : (1)
Solution:
Goblet cells of the alimentary canal secrete mucus.

(141) Answer : (4)
Solution:
In heart, kidney and capillaries cartilage is absent.

(142) Answer : (1)
Hint:
Alternate arrangement of dark and light bands
Solution:
Skeletal muscle fibres are unbranched, striated, voluntary and devoid of communication junctions. Cardiac muscle fibres are striated, branched, involuntary and contain communication junctions at some fusion points (intercalated discs).

(143) Answer : (1)
Hint:
They form the main skeletal framework of body.
Solution:
Bones interact with skeletal muscles having cylindrical fibres to bring about movements.
Bone marrow in some bones is the site of production of blood cells.

(144) Answer : (1)
Solution:
Cell junctions fuse the plasma membranes of cardiac muscle cells and make them stick together.

(145) Answer : (4)
Solution:
Neuroglial cells support and protect neurons.

(146) Answer : (3)
Hint:
Compound epithelium
Solution:
Compound epithelium is made of more than one layer (multi-layer) of cells and thus has a limited role in secretion and absorption. Their main function is to provide protection against chemical and mechanical stresses. They cover the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts.
Cuboidal epithelium has cube-like cells with central nuclei.
Columnar epithelium has tall and slender cells with basal nuclei.

(147) Answer : (2)
Hint:
Contraction leads to shortening
Solution:
Muscle fibres contract (shorten) in response to stimulation, then relax (lengthen) and return to their uncontracted state in a coordinated fashion.

(148) Answer : (1)
Hint:
Blood is fibreless
Solution:
The defining characteristic of connective tissue is the presence of a large amount of extracellular matrix (ECM). This matrix consists of ground substance (which can be fluid, gel-like, or solid) and fibres (such as collagen, elastin, or reticular fibres). The ECM provides support and structure to tissues and organs.
Blood does not contain fibres.

(149) Answer : (4)
Hint:
Cylindrical cells.
Solution:
Striated muscle fibres/cells have peripheral nuclei. Central nucleus is present in the cells of cuboidal epithelium, cardiac and smooth muscle cells.

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(150) Answer : (3)

Solution:

Smooth muscles, cardiac muscles and epithelial tissue, they all have the presence of cell junctions in them.
Three types of cell junctions are found in epithelial and other tissues.
Cell junctions hold smooth muscle fibres together and they are bundled together in a connective tissue sheath.
Cell junctions fuse the plasma membrane of cardiac muscle cells and make them stick together.

(151) Answer : (2)

Solution:

Tight junctions (Zona occludens) help to stop substances from leaking across the tissue.

(152) Answer : (1)

Solution:

Fallopian tubes (oviducts) are lined by ciliated columnar epithelium.

(153) Answer : (3)

Hint:

Dense regular connective tissue

Solution:

In dense connective tissue, orientation of fibres show a regular or irregular pattern and are called dense regular and dense irregular tissues respectively. In the dense regular connective tissues, the collagen fibres are present in rows between many parallel bundles of fibres. Tendons, which attach skeletal muscles to bones and ligaments which attach one bone to another are examples of this tissue.

(154) Answer : (4)

Solution:

Connective tissue is the most abundant and widely distributed tissue in the body of complex animals. It includes bone, cartilage, adipose tissue, areolar tissue and blood.

(155) Answer : (1)

Hint:

Multicellular animal

Solution:

In unicellular organisms, all functions like digestion, respiration and reproduction are performed by a single cell. In the complex body of multicellular animals, the same basic functions are carried out by different groups of cells in a well organised manner. The body of a simple organism like *Hydra* is made of different types of cells and the number of cells in each type can be in thousands. *Hydra* exhibits division of labour. Others are unicellular organisms which do not show division of labour at tissue level.

(156) Answer : (4)

Solution:

Tissues are organised in specific proportions and patterns to form an organ.
All types of tissues are present in organs such as stomach, heart, lung and kidney.

(157) Answer : (1)

Hint:

Collagen or elastin fibres

Solution:

In all connective tissues except blood, the fibroblasts secrete fibres of structural proteins called collagen or elastin. The fibres provide strength, elasticity and flexibility to the tissue.

(158) Answer : (1)

Hint:

Water is the most abundant chemical in living tissue.

Solution:

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

(159) Answer : (2)

Solution:

Hydroxymethyl is present as side chain in serine.

(160) Answer : (3)

Hint:

DNA possesses deoxyribose sugar.

Solution:

The sugar found in polynucleotides is either ribose (a monosaccharide pentose) or 2' deoxyribose. A nucleic acid containing deoxyribose sugar is called deoxyribonucleic acid (DNA) while the one which contains ribose sugar is called ribonucleic acid.

(161) Answer : (4)

Solution:

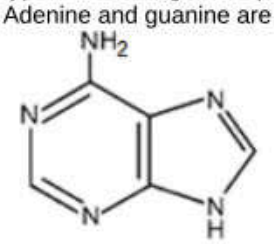
Collagen is the most abundant protein in the animal world and Ribulose Bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the whole of the biosphere.

(162) Answer : (3)

Solution:

Trypsin and collagen are proteins (heteropolymers of amino acids).

• Adenine and guanine are substituted purines. They contain two heterocyclic rings in their structure.



Adenine (nitrogenous base)

• Oils have lower melting point (*e.g.*, Gingelly oil) and hence remain as oil in winters. Lipids are also called fats and oils based on their melting points.

• Exoskeleton of arthropods have a complex polysaccharide called chitin. The monomeric unit of chitin is N-acetyl glucosamine (NAG).

(163) Answer : (3)

Solution:

A competitive inhibitor binds reversibly with the enzyme. Its effect can be overcome by increasing the concentration of the substrate.

Competitive inhibition refers to a situation where an inhibitor competes with the substrate for binding at the substrate binding site of an enzyme. It increases the K_m value but does not affect the maximum velocity (V_{max}) of the reaction.

(164) Answer : (2)

Solution:

Adult human haemoglobin consists of 4 subunits. Two of these are identical to each other. Hence, two subunits of 'α' type and two subunits of 'β' type together constitute the human haemoglobin (Hb).

(165) Answer : (3)

Hint:

Property of zwitterion

Solution:

Lysine is a basic amino acid while valine is a neutral amino acid.

Basic amino acids have an additional amino group. Neutral amino acids have equal number of amino group and carboxyl group.

(166) Answer : (1)

Solution:

Uridine is a nucleoside while uridylic acid is a nucleotide.

(167) Answer : (2)

Hint:

Co-factor = Prosthetic groups, co-enzymes and metal ions

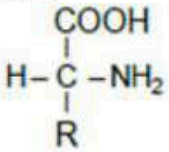
Solution:

Haem is the prosthetic group of enzymes 'peroxidase' and 'catalase'.

A prosthetic group is an organic compound and is always tightly bound to an enzyme.

(168) Answer : (4)

Hint:



Solution:

Amino acids are substituted methanes. There are four substituent groups occupying the four valency positions. These are hydrogen, carboxyl group, amino group and a variable group designated as R group.

(169) Answer : (2)

Hint:

Storage polysaccharide in plants

Solution:

Glucose is a monosaccharide.
Cellulose does not contain complex helices and hence cannot hold I_2.
Glycogen gives reddish-brown colour with iodine.
Starch can hold I_2 molecules in the helical portion and gives blue colour.

(170) Answer : (4)
Solution:
K_m represents the concentration of substrate at $\frac{V_{max}}{2}$.

(171) Answer : (4)
Solution:
Enzyme-substrate complex passes through transition state to produce product and enzyme again becomes free.

(172) Answer : (1)
Solution:
Rate doubles or decreases by half for every 10°C change in either direction.

(173) Answer : (2)
Solution:
Each enzyme (E) has a substrate (S) binding site in its molecule so that a highly reactive enzyme-substrate complex (ES) is produced. This complex is short-lived and dissociates into its product(s) 'P' and the unchanged enzyme with an intermediate formation of the enzyme-product complex (EP).

(174) Answer : (3)
Solution:
Enzymes are divided into 6 classes each with 4-13 sub classes and named accordingly by a four digit number. Lyases belong to class IV of enzymes.

(175) Answer : (4)
Solution:
Steroids are derived lipids.

(176) Answer : (3)
Solution:
In a double stranded DNA molecule, the amount of adenine will be equal to the amount of thymine and the amount of guanine will be equal to the amount of cytosine.
Given, A = 32%, hence, T = 32%.
A + T = 64%. Hence, G + C = 100% – 64% = 36%.
$\therefore$ G = C $\because$ G = 18%, C = 18%.

(177) Answer : (2)
Hint:
Palmitic acid is a saturated fatty acid.
Solution:
Palmitic acid has 16 carbons including the carboxyl carbon. Therefore, correct representation of palmitic acid is $CH_3 – (CH_2)_{14} – COOH$.

(178) Answer : (3)
Solution:
Glycosidic bond is formed between the sugar molecule and nitrogenous base molecule.
Hydrogen bond is present between the bases in opposing strands of DNA. The bond between the hydroxyl group of sugar and phosphate is ester bond.

(179) Answer : (1)
Solution:
Change in factors such as temperature, pH, change in substrate concentration or binding of specific chemical and co-factor will alter the enzymatic activity of holoenzyme.

(180) Answer : (3)
Solution:
NAD and NADP contain the vitamin niacin.

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